Week 7: Kinship (Lab answers)

SOC6708 ADA

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In this lab we will be calculating surviving daughters and mothers using the GKP equations, and also exploring matrix-based models using `DemoKin`.

Load in data

Note you will have to install the DemoKin package.

```
library(tidyverse)
library(readxl)
library(DemoKin)
```

For the GKP we will be using fertility and mortality rates from WPP; we also need to load in the file that has summary indicators. You should have these files from previous weeks.

```
df <- read_csv("/Users/liangqishen/Desktop/WPP2024_Fertility_by_Age5.csv")
dm <- read_csv("/Users/liangqishen/Desktop/WPP2024_Life_Table_Abridged_Medium_1950-2023.csv")
di <- read_csv("/Users/liangqishen/Desktop/WPP2024_Demographic_Indicators_Medium.csv")
```

For the matrix models, we will use fertility and mortality rates by single year of age. I've already cleaned these up (the file is in the repo).

```
ds <- read_csv("/Users/liangqishen/Desktop/wpp_pop_fx_px_female.csv")
```

Goodman-Keyfitz-Pullum equations

Surviving daughters

Let's look at the surviving daughters in Kenya over time.

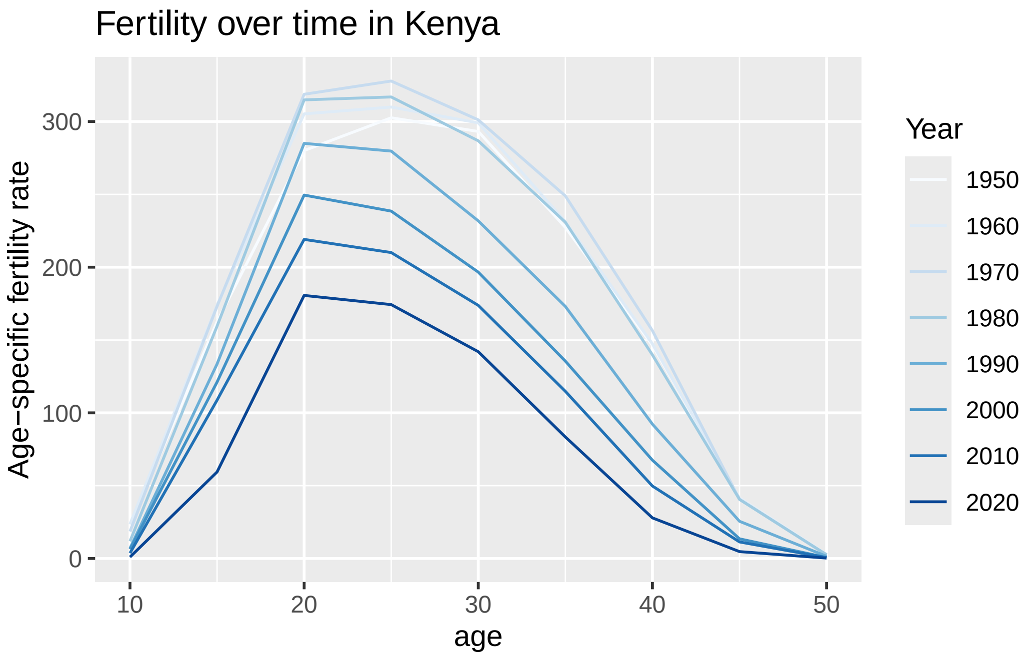
First, clean up the fertility and mortality data:

```
fert <- df |>
  filter(Location == "Kenya") |>
  select(Time, AgeGrp, ASFR) |>
  mutate(age = as.numeric(str_remove(AgeGrp, "-.*")))

mort <- dm |>
  filter(Location == "Kenya", Sex == "Female") |>
  select(Time, AgeGrp, lx) |>
  filter(AgeGrp!="1-4")|> # don't need this
  mutate(age = as.numeric(str_remove(AgeGrp, "-.*")))
```

Exercise

Do a couple of graphs to show how fertility and mortality have changed over time.

```
fert |>
  filter(Time %in% seq(1950, 2020, by = 10)) |>
  ggplot(aes(age, ASFR, color = factor(Time))) +
  geom_line()+
  scale_color_brewer(name = "Year")+
  labs(y = "Age-specific fertility rate", title = "Fertility over time in Kenya")
```



Let's calculate expected number of surviving daughters over time for a 60 year old:

```
a <- 60

surviving_daughters <- fert |>
  mutate(shifted_age = a - age) |>
  left_join(mort |> rename(shifted_age = age) |>
              select(-AgeGrp) ) |>
  mutate(prod = ASFR*5*0.4886/1000*lx/100000) |>
  group_by(Time) |>
  summarize(daughters = sum(prod)) |>
  filter(Time < 2024)
```

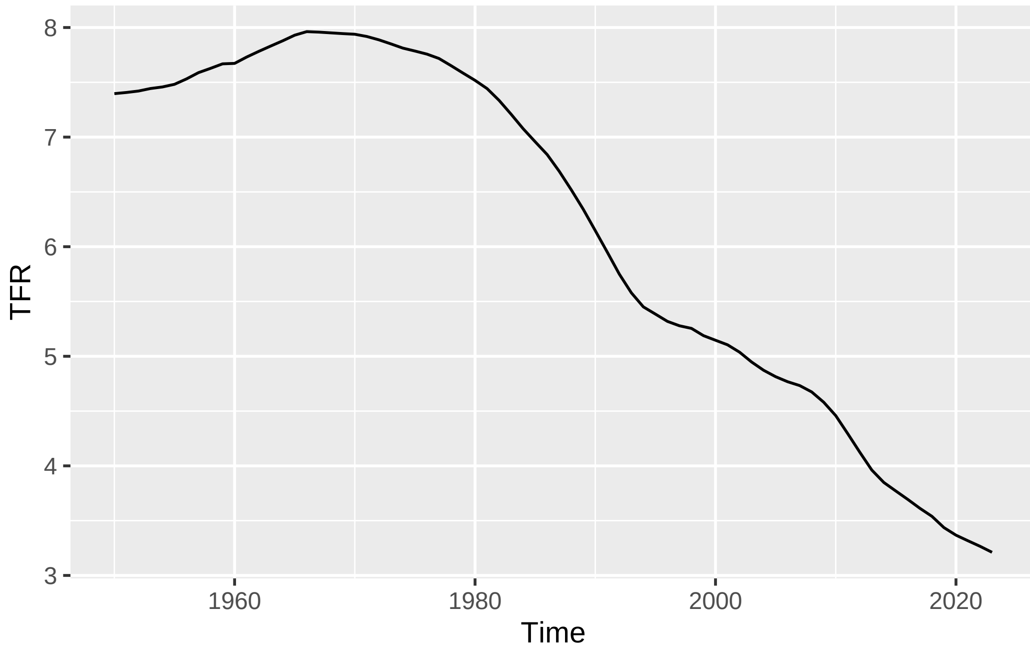
Plot this:

```
surviving_daughters |>
  ggplot(aes(Time, daughters)) +
  geom_line()+
  labs(y = "Surviving daughters", title = "Expected number of surviving daughters for a 60 ye
```

Exercise

Compare the trajectory above to the trajectory of TFR for Kenya. Explain the similarities / differences.

```
fert |>
  group_by(Time) |>
  summarize(TFR = sum(ASFR*5/1000)) |>
  filter(Time < 2024) |>
  ggplot(aes(Time, TFR))+
  geom_line()
```



Surviving mothers

To calculate the probability that a mother is still alive, we need to calculate W_x, which is the age distribution of mothers. We'll calculate this using stable population assumptions as in the lecture notes, which requires first calculating the stable growth rate r. I'm using the approximation from week 5, i.e.

$$r \approx \frac{NRR}{\mu}$$

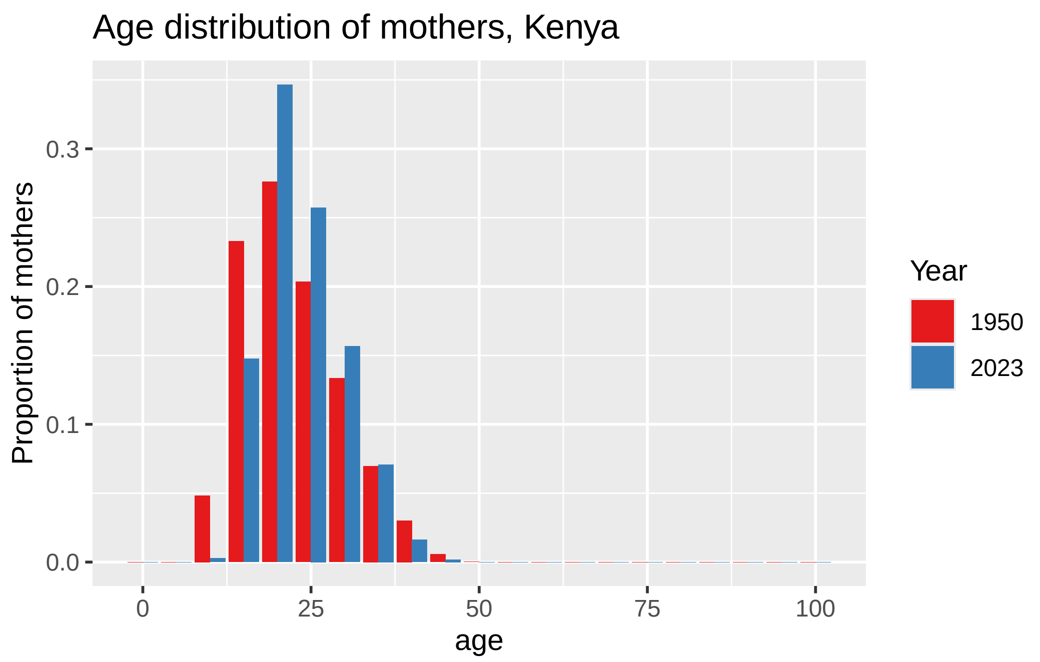
where μ is the mean age at childbearing.

```
dr <- di |>
  filter(Location=="Kenya") |>
  select(Time, NRR, MAC) |>
  mutate(r = NRR/MAC)

Wx <- mort |>
  left_join(fert) |>
  mutate(ASFR = ifelse(is.na(ASFR), 0, ASFR)) |>
  mutate(age = ifelse(is.na(age), 100, age)) |>
  mutate(prod = ASFR*5*0.4886/1000*lx/100000) |>
  left_join(dr) |>
```

```r
  mutate(dist = exp(-1*r*age)*prod) |>
  group_by(Time) |>
  mutate(sum = sum(dist)) |>
  mutate(prop = dist/sum) # renormalize
```

Plot a couple of years to make sure it broadly makes sense

```r
Wx |>
  filter(Time == 2023|Time == 1950) |>
  ggplot(aes(age, prop, fill = factor(Time))) +
  geom_bar(stat = "identity", position = 'dodge')+
  scale_fill_brewer(palette = "Set1", name = "Year")+
  labs(y = "Proportion of mothers", title = "Age distribution of mothers, Kenya")
```

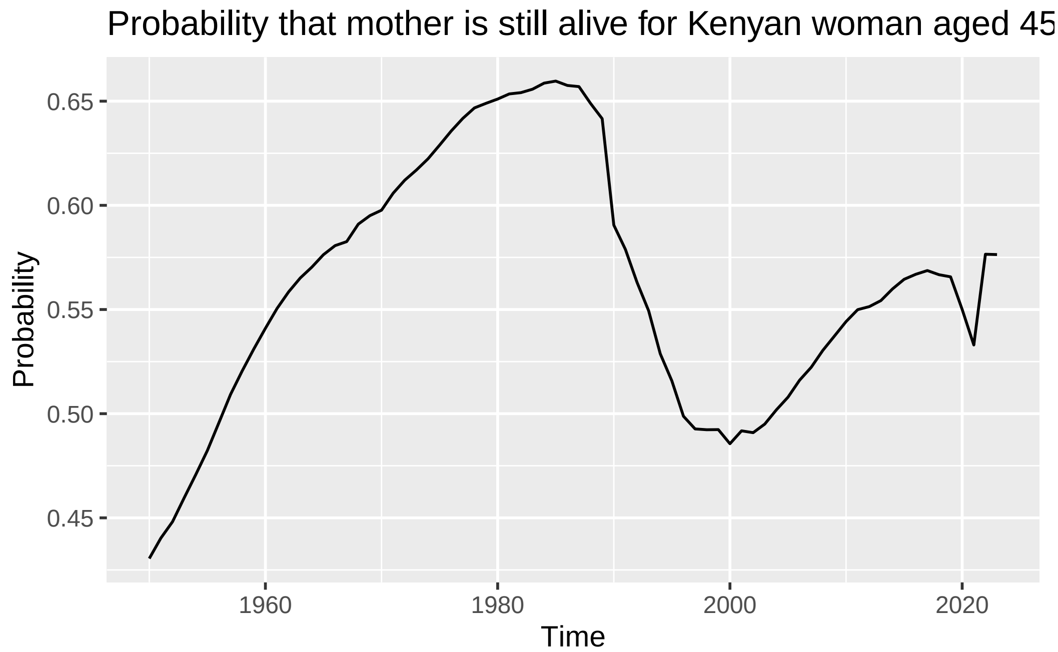


Now we have this age distribution, we can calculate the probability that a mother is alive. For this one, let's use ego age = 45.

```r
a <- 45

age_shifted = seq(0, 100, by = 5) + a

# shift mortality rates
```

```r
mort_shift <- mort |>
  mutate(age = ifelse(is.na(age), 100, age)) |>
  filter(age %in% age_shifted) |>
  rename(lx_shifted = lx, age_shifted = age) |>
  select(-AgeGrp)

surviving_mothers <- mort |>
  mutate(age_shifted = age + a) |>
  left_join(mort_shift) |>
  mutate(lx_shifted = ifelse(is.na(lx_shifted), 0, lx_shifted)) |>
  left_join(fert) |>
  mutate(ASFR = ifelse(is.na(ASFR), 0, ASFR)) |>
  left_join(Wx |>
              select(Time, AgeGrp, age, prop)) |>
  mutate(ratio = lx_shifted/lx) |>
  mutate(prod = prop*ratio) |>
  drop_na(age) |>
  group_by(Time) |>
  summarize(M1 = sum(prod))
```

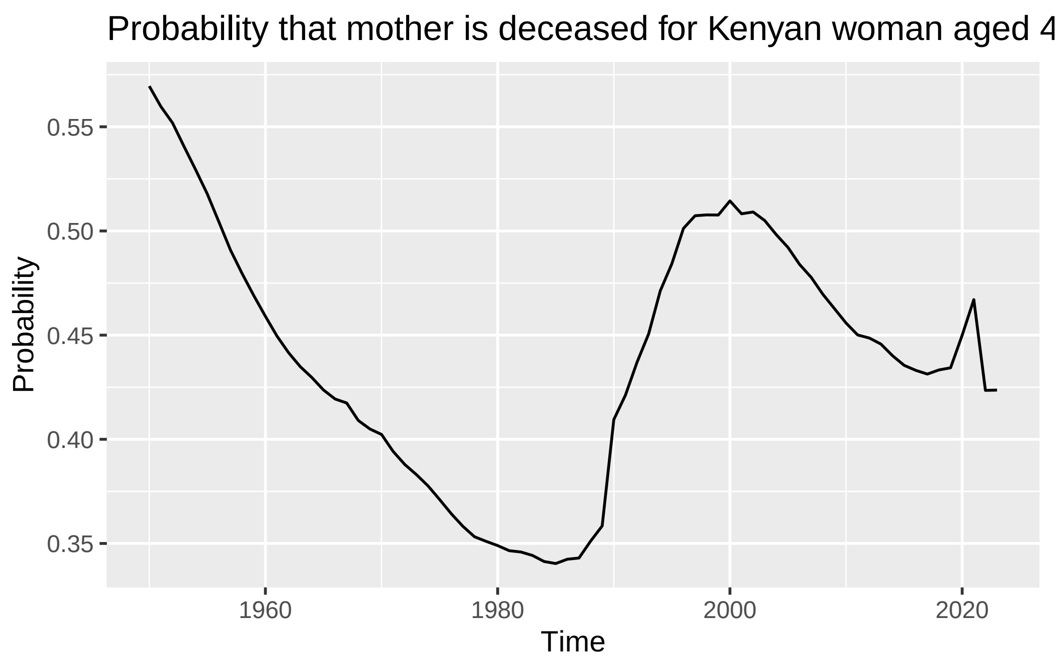
```r
surviving_mothers|>
  ggplot(aes(Time, M1)) +
  geom_line()+
  labs(y = "Probability", title = "Probability that mother is still alive for Kenyan woman ag
```



Question for you

How would you convert this probability into the number of 45 year old women in Kenya who have lost their mother (say, in 2023)?

```
no_surviving_mothers <- surviving_mothers |>
   mutate (M2 = 1-M1)

no_surviving_mothers |>
  ggplot(aes(Time, M2)) +
  geom_line()+
  labs(y = "Probability", title = "Probability that mother is deceased for Kenyan woman aged
```



DemoKin

Now we will use the `DemoKin` package to calculate surviving kin of different types (female only) in Canada. This code replicates the plots in the lecture notes.

Let's get the fertility and mortality rates in the right format:

```
asfr_2023 <- ds |>
  filter(region=="Canada") |>
  filter(year==2023) |>
  select(age, year, fx) |>
  mutate(fx = fx/1000) |>
  pivot_wider(names_from = year, values_from = fx) |>
  select(-age) |>
  as.matrix()

surv_2023 <- ds |>
  filter(region=="Canada") |>
  filter(year==2023) |>
  select(age, year, px) |>
  pivot_wider(names_from = year, values_from = px) |>
```

```
  select(-age) |>
  as.matrix()
```

The main function in the DemoKin package is then `kin`, which takes an inputs fertility and survival matrices.

```
kin_2023 <- kin(p = surv_2023, f = asfr_2023, time_invariant = TRUE)
```

This object has two elements, a full description and a summary.

```
kin_2023$kin_full
```

```
# A tibble: 142,814 x 7
   kin   age_kin age_focal living  dead cohort year 
   <chr>   <int>     <int>  <dbl> <dbl> <lgl>  <lgl>
 1 d           0         0      0     0 NA     NA   
 2 d           0         1      0     0 NA     NA   
 3 d           0         2      0     0 NA     NA   
 4 d           0         3      0     0 NA     NA   
 5 d           0         4      0     0 NA     NA   
 6 d           0         5      0     0 NA     NA   
 7 d           0         6      0     0 NA     NA   
 8 d           0         7      0     0 NA     NA   
 9 d           0         8      0     0 NA     NA   
10 d           0         9      0     0 NA     NA   
# i 142,804 more rows
```

```
unique(kin_2023$kin_full[,"kin"])
```

```
# A tibble: 14 x 1
   kin  
   <chr>
 1 d    
 2 gd   
 3 ggd  
 4 m    
 5 gm   
 6 ggm  
 7 os   
 8 ys   
```

```
 9 nos
10 nys
11 oa
12 ya
13 coa
14 cya
```

```r
kin_2023$kin_summary
```

```
# A tibble: 1,414 x 9
   age_focal kin   year  count_living mean_age sd_age  count_dead count_cum_dead
       <int> <chr> <lgl>        <dbl>    <dbl>  <dbl>       <dbl>          <dbl>
 1         0 coa   NA       0.153         8.59   6.33 0.0000513      0.0000513
 2         0 cya   NA       0.0491        4.21   3.84 0.0000333      0.0000333
 3         0 d     NA       0           NaN    NaN    0              0
 4         0 gd    NA       0           NaN    NaN    0              0
 5         0 ggd   NA       0           NaN    NaN    0              0
 6         0 ggm   NA       0.301        86.2    6.66 0.0274         0.0274
 7         0 gm    NA       0.934        62.2    7.26 0.00613        0.00613
 8         0 m     NA       1            31.3    5.20 0.000500       0.000500
 9         0 nos   NA       0.0000656     1.45   1.75 0.000000102    0.000000102
10         0 nys   NA       0           NaN    NaN    0              0
# i 1,404 more rows
# i 1 more variable: mean_age_lost <dbl>
```

Note that this kin column refers to different types of kin, to make this more readable we can use the `rename_kin` function:

```r
kin_2023$kin_summary |>
  rename_kin()
```

```
# A tibble: 1,414 x 10
   age_focal kin   year  count_living mean_age sd_age  count_dead count_cum_dead
       <int> <chr> <lgl>        <dbl>    <dbl>  <dbl>       <dbl>          <dbl>
 1         0 coa   NA       0.153         8.59   6.33 0.0000513      0.0000513
 2         0 cya   NA       0.0491        4.21   3.84 0.0000333      0.0000333
 3         0 d     NA       0           NaN    NaN    0              0
 4         0 gd    NA       0           NaN    NaN    0              0
 5         0 ggd   NA       0           NaN    NaN    0              0
 6         0 ggm   NA       0.301        86.2    6.66 0.0274         0.0274
 7         0 gm    NA       0.934        62.2    7.26 0.00613        0.00613
```

```
 8          0 m      NA       1             31.3    5.20 0.000500        0.000500
 9          0 nos    NA       0.0000656      1.45   1.75 0.000000102     0.000000102
10          0 nys    NA       0            NaN     NaN    0               0
# i 1,404 more rows
# i 2 more variables: mean_age_lost <dbl>, kin_label <chr>
```

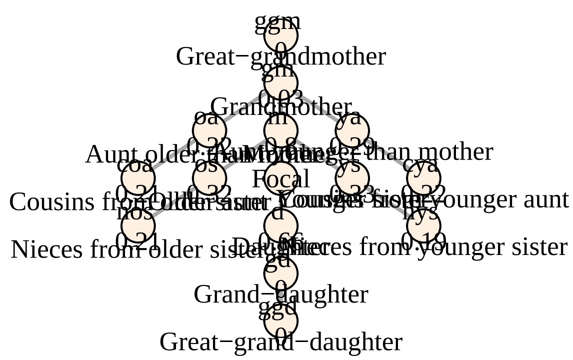
For example, let's look at the number of surviving daughters to 60 year olds:

```
kin_2023$kin_summary |>
  rename_kin() |>
  filter(kin_label == "Daughters", age_focal==60)
```

```
# A tibble: 1 x 10
  age_focal kin   year  count_living mean_age sd_age count_dead count_cum_dead
      <int> <chr> <lgl>        <dbl>    <dbl>  <dbl>      <dbl>          <dbl>
1        60 d     NA           0.655     28.1   5.21   0.000277        0.00627
# i 2 more variables: mean_age_lost <dbl>, kin_label <chr>
```

We can plot a tree summary for a 45 year old:

```
kin_2023$kin_summary |>
  filter(age_focal == 45) |>
  select(kin, count = count_living) |>
  plot_diagram(rounding = 2)
```

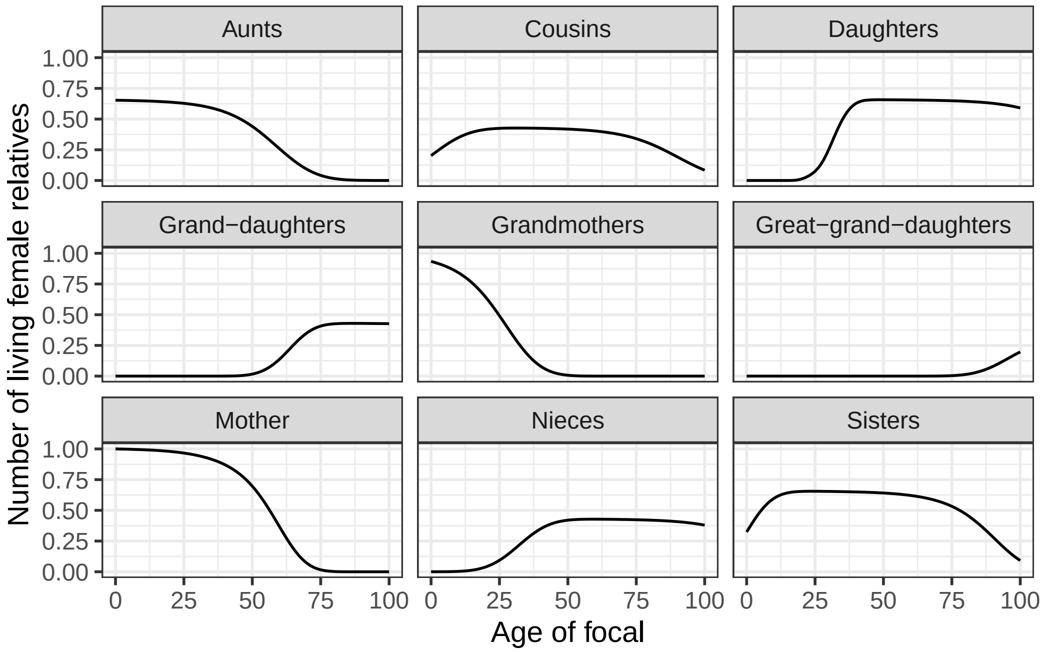


We can specify the types of kin we are interested in.

```
kin_2023 <-
  kin(
    p = surv_2023
    , f = asfr_2023
    , output_kin = c("c", "d", "gd", "ggd", "gm", "m", "n", "a", "s")
    , time_invariant = TRUE
  )
```

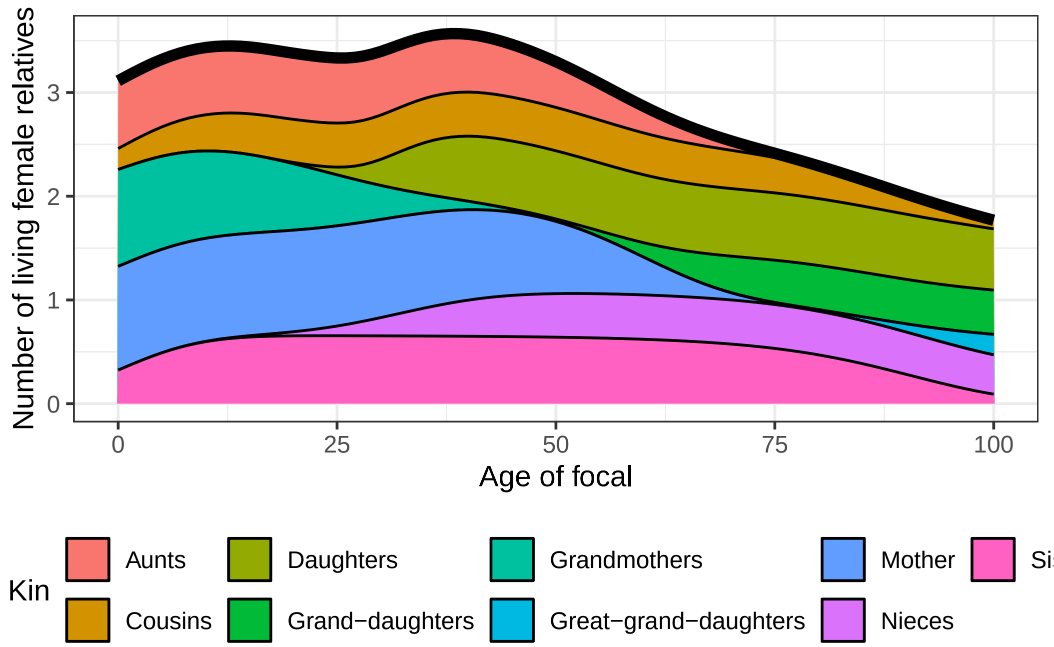
Then plot the number surviving over age

```
kin_2023$kin_summary |>
  rename_kin() |>
  ggplot() +
  geom_line(aes(age_focal, count_living))  +
  theme_bw() +
  labs(x = "Age of focal", y= "Number of living female relatives") +
  facet_wrap(~kin_label)
```



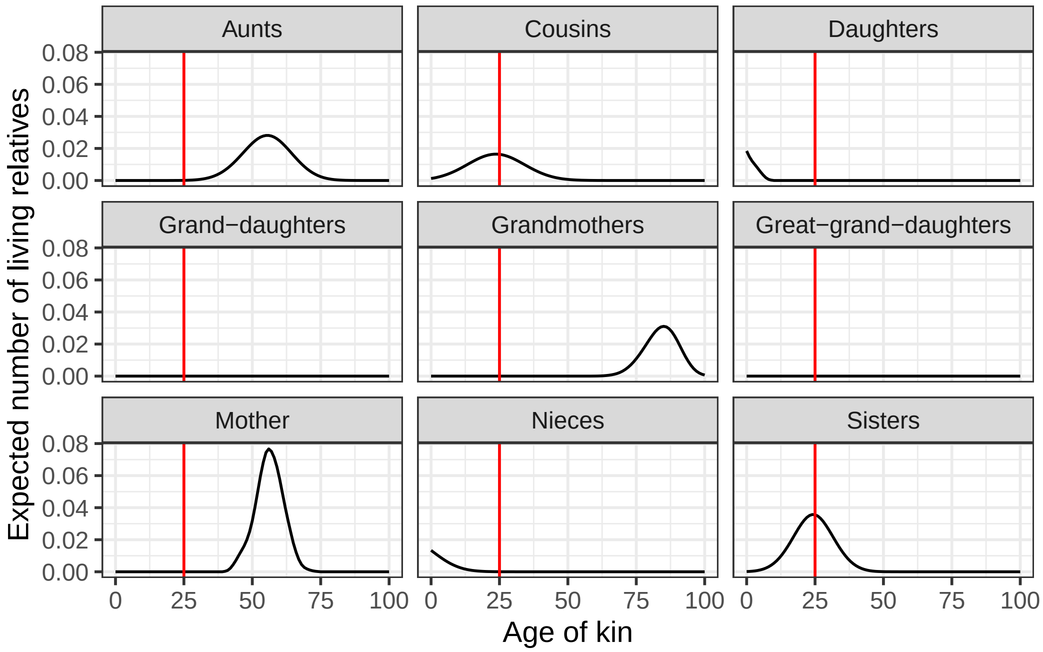
Combine these counts:

```
counts <-
  kin_2023$kin_summary |>
  group_by(age_focal) |>
  summarise(count_living = sum(count_living)) |>
  ungroup()

kin_2023$kin_summary |>
  select(age_focal, kin, count_living) |>
  rename_kin() |>
  ggplot(aes(x = age_focal, y = count_living)) +
  geom_area(aes(fill = kin_label), colour = "black") +
  geom_line(data = counts, linewidth = 2) +
  labs(x = "Age of focal",
       y = "Number of living female relatives",
       fill = "Kin") +
  theme_bw() +
  theme(legend.position = "bottom")
```



Finally, we can plot the age distribution of relatives for Canadian women aged 65.

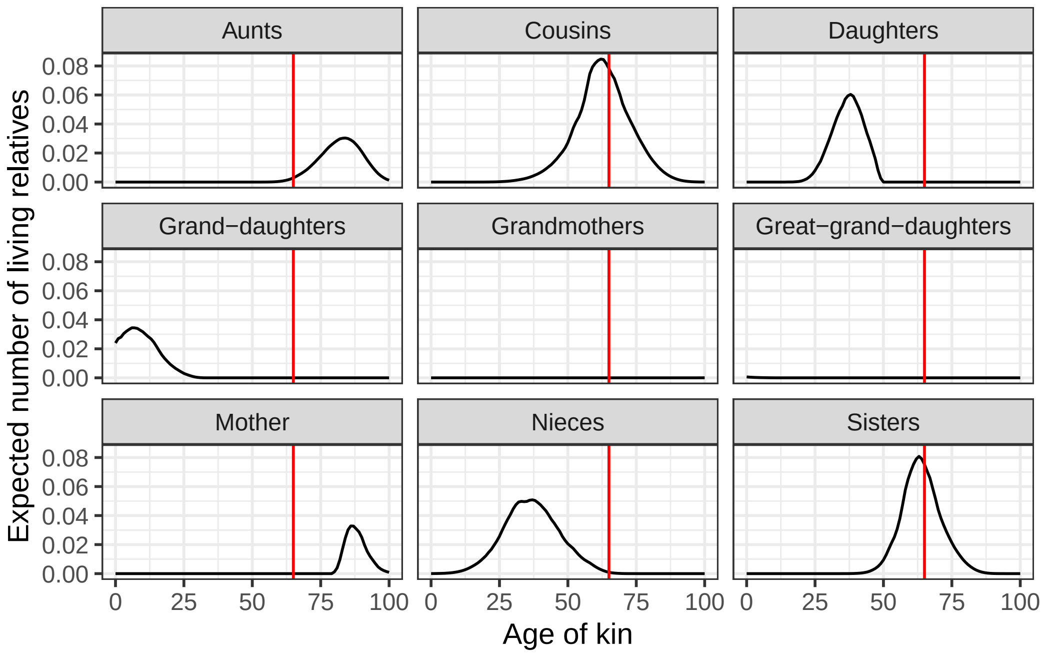
```
kin_2023$kin_full |>
  rename_kin() |>
  filter(age_focal == 25) |>
  ggplot(aes(age_kin, living)) +
  geom_line() +
  geom_vline(xintercept = 25, colour = "red") +
  labs(x = "Age of kin",
       y = "Expected number of living relatives") +
  theme_bw() +
  facet_wrap(~kin_label)
```



Can repeat the same exercise using time varying rates:

```
asfr <- ds |>
  filter(region=="Canada") |>
  select(age, year, fx) |>
  mutate(fx = fx/1000) |>
  pivot_wider(names_from = year, values_from = fx) |>
  select(-age) |>
  as.matrix()

surv <- ds |>
  filter(region=="Canada") |>
  select(age, year, px) |>
  pivot_wider(names_from = year, values_from = px) |>
  select(-age) |>
  as.matrix()

kin_tv <- kin(p = surv, f = asfr, time_invariant = FALSE,
              output_kin = c("c", "d", "gd", "ggd", "gm", "m", "n", "a", "s"))

kin_tv$kin_full |>
  filter(year==2023) |>
  rename_kin() |>
```

```
  filter(age_focal == 65) |>
  ggplot(aes(age_kin, living)) +
  geom_line() +
  geom_vline(xintercept = 65, colour = "red") +
  labs(x = "Age of kin",
       y = "Expected number of living relatives") +
  theme_bw() +
  facet_wrap(~kin_label)
```



Exercise

Pick another country to get the kinship structure for women. Compare to Canada and briefly discuss similarities and differences.

```
asfr_2023 <- ds |>
  filter(region=="China") |>
  filter(year==2023) |>
  select(age, year, fx) |>
  mutate(fx = fx/1000) |>
  pivot_wider(names_from = year, values_from = fx) |>
  select(-age) |>
```

```
  as.matrix()

surv_2023 <- ds |>
  filter(region=="China") |>
  filter(year==2023) |>
  select(age, year, px) |>
  pivot_wider(names_from = year, values_from = px) |>
  select(-age) |>
  as.matrix()

kin_2023 <- kin(p = surv_2023, f = asfr_2023, time_invariant = TRUE)

kin_2023$kin_full
```

```
# A tibble: 142,814 x 7
   kin   age_kin age_focal living  dead cohort year
   <chr>   <int>     <int>  <dbl> <dbl> <lgl>  <lgl>
 1 d           0         0      0     0 NA     NA
 2 d           0         1      0     0 NA     NA
 3 d           0         2      0     0 NA     NA
 4 d           0         3      0     0 NA     NA
 5 d           0         4      0     0 NA     NA
 6 d           0         5      0     0 NA     NA
 7 d           0         6      0     0 NA     NA
 8 d           0         7      0     0 NA     NA
 9 d           0         8      0     0 NA     NA
10 d           0         9      0     0 NA     NA
# i 142,804 more rows
```

```
unique(kin_2023$kin_full[,"kin"])
```

```
# A tibble: 14 x 1
   kin
   <chr>
 1 d
 2 gd
 3 ggd
 4 m
 5 gm
 6 ggm
 7 os
```

```
 8 ys
 9 nos
10 nys
11 oa
12 ya
13 coa
14 cya
```

```r
kin_2023$kin_summary
```

```
# A tibble: 1,414 x 9
   age_focal kin   year  count_living mean_age sd_age  count_dead count_cum_dead
       <int> <chr> <lgl>        <dbl>    <dbl>  <dbl>       <dbl>          <dbl>
 1         0 coa   NA        0.0897       9.77   7.13 0.0000470      0.0000470
 2         0 cya   NA        0.0280       4.82   4.37 0.0000268      0.0000268
 3         0 d     NA        0          NaN    NaN    0              0
 4         0 gd    NA        0          NaN    NaN    0              0
 5         0 ggd   NA        0          NaN    NaN    0              0
 6         0 ggm   NA        0.375       80.5    6.75 0.0277         0.0277
 7         0 gm    NA        0.938       58.1    7.91 0.00618        0.00618
 8         0 m     NA        1           29.3    5.80 0.000582       0.000582
 9         0 nos   NA        0.000236     1.72   1.89 0.000000447    0.000000447
10         0 nys   NA        0          NaN    NaN    0              0
# i 1,404 more rows
# i 1 more variable: mean_age_lost <dbl>
```

```r
kin_2023$kin_summary |>
  rename_kin()
```

```
# A tibble: 1,414 x 10
   age_focal kin   year  count_living mean_age sd_age  count_dead count_cum_dead
       <int> <chr> <lgl>        <dbl>    <dbl>  <dbl>       <dbl>          <dbl>
 1         0 coa   NA        0.0897       9.77   7.13 0.0000470      0.0000470
 2         0 cya   NA        0.0280       4.82   4.37 0.0000268      0.0000268
 3         0 d     NA        0          NaN    NaN    0              0
 4         0 gd    NA        0          NaN    NaN    0              0
 5         0 ggd   NA        0          NaN    NaN    0              0
 6         0 ggm   NA        0.375       80.5    6.75 0.0277         0.0277
 7         0 gm    NA        0.938       58.1    7.91 0.00618        0.00618
 8         0 m     NA        1           29.3    5.80 0.000582       0.000582
 9         0 nos   NA        0.000236     1.72   1.89 0.000000447    0.000000447
```

```
10          0 nys    NA         0           NaN     NaN     0               0
# i 1,404 more rows
# i 2 more variables: mean_age_lost <dbl>, kin_label <chr>
```

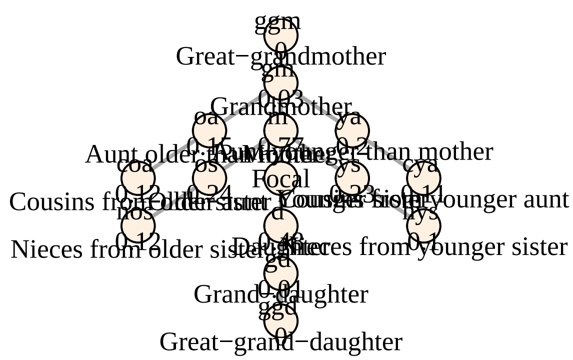
surviving daughters to 60 year olds: Compared to 60-year-old Canadian women, 60-year-old Chinese women are less likely to have a surviving daughters.

```
kin_2023$kin_summary |>
  rename_kin() |>
  filter(kin_label == "Daughters", age_focal==60)
```

```
# A tibble: 1 x 10
  age_focal kin   year  count_living mean_age sd_age count_dead count_cum_dead
      <int> <chr> <lgl>        <dbl>    <dbl>  <dbl>      <dbl>          <dbl>
1        60 d     NA           0.482     30.5   5.60   0.000288        0.00769
# i 2 more variables: mean_age_lost <dbl>, kin_label <chr>
```

45-year-old Chinese woman, kinship tree compared to 45-year-old canadian women, 45-year-old Chinese women are less likely to have almost all types of kin

```
kin_2023$kin_summary |>
  filter(age_focal == 45) |>
  select(kin, count = count_living) |>
  plot_diagram(rounding = 2)
```

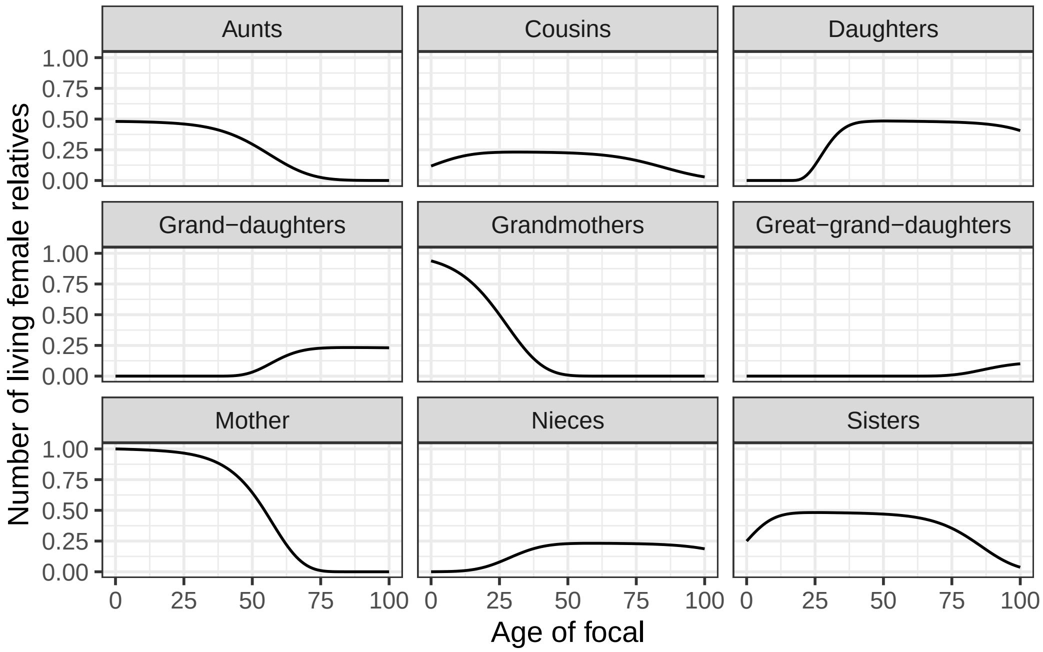


We can specify the types of kin we are interested in.

```
kin_2023 <-
  kin(
    p = surv_2023
    , f = asfr_2023
    , output_kin = c("c", "d", "gd", "ggd", "gm", "m", "n", "a", "s")
    , time_invariant = TRUE
  )
```

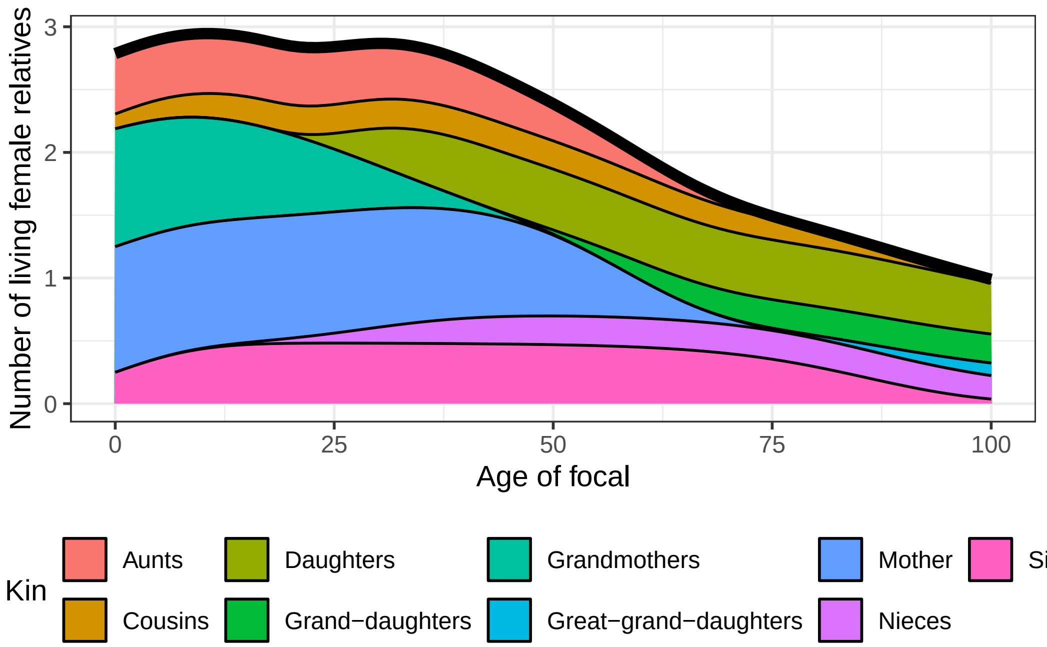
Then plot the number surviving over age

```
kin_2023$kin_summary |>
  rename_kin() |>
  ggplot() +
  geom_line(aes(age_focal, count_living))  +
  theme_bw() +
  labs(x = "Age of focal", y= "Number of living female relatives") +
  facet_wrap(~kin_label)
```



Combine these counts:

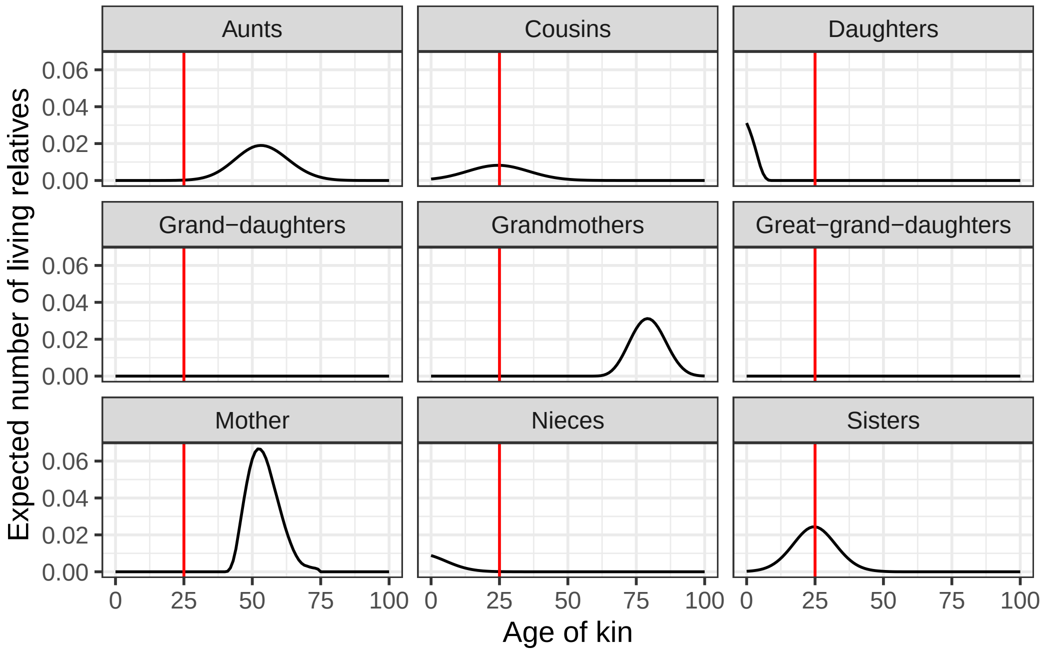
```
counts <-
  kin_2023$kin_summary |>
  group_by(age_focal) |>
  summarise(count_living = sum(count_living)) |>
  ungroup()

kin_2023$kin_summary |>
  select(age_focal, kin, count_living) |>
  rename_kin() |>
  ggplot(aes(x = age_focal, y = count_living)) +
  geom_area(aes(fill = kin_label), colour = "black") +
  geom_line(data = counts, linewidth = 2) +
  labs(x = "Age of focal",
       y = "Number of living female relatives",
       fill = "Kin") +
  theme_bw() +
  theme(legend.position = "bottom")
```



less aunts, less cousins, less daughter, less sisters; almost less on everything, except for mothers and grandmothers.

we can plot the age distribution of relatives for Chinese women aged 65.

```
kin_2023$kin_full |>
  rename_kin() |>
  filter(age_focal == 25) |>
  ggplot(aes(age_kin, living)) +
  geom_line() +
  geom_vline(xintercept = 25, colour = "red") +
  labs(x = "Age of kin",
       y = "Expected number of living relatives") +
  theme_bw() +
  facet_wrap(~kin_label)
```


Aunts
Cousins
Daughters
Grand−daughters
Grandmothers
Great−grand−daughters
Mother
Nieces
Sisters
Expected number of living relatives
Age of kin
0.06
0.04
0.02
0.00
0
25
50
75
100